

C# Library System — Windows Forms

Step-by-Step Instructional Guide for Claude Code

This guide covers building a 4-form Windows Forms application named **LibrarySystem**. The project includes a Login screen, a main Menu with MenuStrip navigation, a Books CRUD form, and a Borrower CRUD form.

Form	Class	Purpose	Key Controls
Form1 (frmLogin)	frmLogin	User login	txtUsername, txtPassword, btnLogin, lblError
Form2	Form2	Main menu	menuStrip1: Master Files, Transaction, Reports, Exit
Form3	Form3	Books CRUD + search	txtBookID/Title/Author, 5 buttons, grid1, txtSearch
Form4	Form4	Borrower CRUD + search	txtBorrowerID/Name/Contact, 5 buttons, grid2, txtSearch

Part 1 — Project Setup

1.1 Create the Project

Open Visual Studio → **Create a new project** → **Windows Forms App (.NET Framework)** → name it **LibrarySystem** → Create.

1.2 Rename Form1 → frmLogin

In Solution Explorer right-click **Form1.cs** → Rename → type **frmLogin.cs**. When prompted, click **Yes** to rename all references.

1.3 Add Remaining Forms

Right-click the project → **Add** → **Windows Form**. Add: **Form2.cs**, **Form3.cs**, **Form4.cs**.

1.4 Fix Startup Form

```
// In Program.cs change: Application.Run(new Form1()); // To: Application.Run(new frmLogin());
```

Part 2 — Form1: frmLogin (Login Screen)

2.1 Sample Screenshot

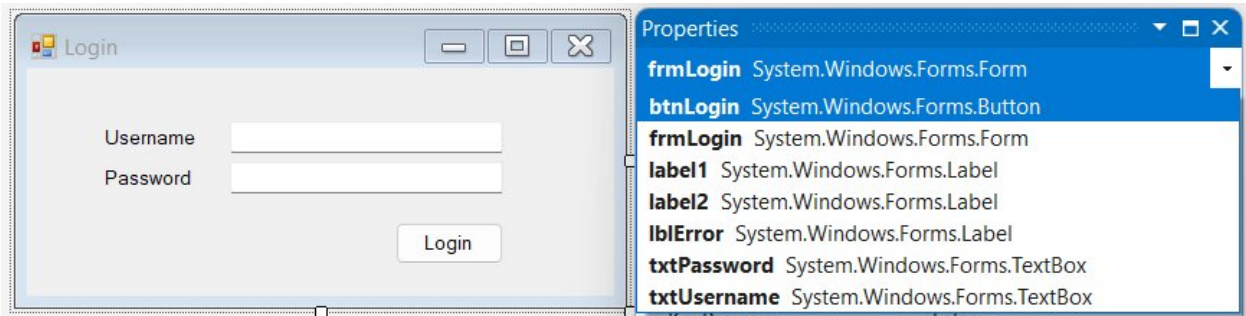


Figure 1 — frmLogin design view with Properties panel showing control names

2.2 Controls to Add

Control Type	Name	Key Settings
Label	label1	Text = "Username"
Label	label2	Text = "Password"
Label	lblError	Text = "" (blank), ForeColor = Red
TextBox	txtUsername	
TextBox	txtPassword	PasswordChar = *
Button	btnLogin	Text = "Login"

2.3 Form Properties

Set **frmLogin.Text** = "Login". Optionally set **FormBorderStyle** = FixedSingle and **MaximizeBox** = False.

2.4 btnLogin Click Code

```
private void btnLogin_Click(object sender, EventArgs e) { string user = txtUsername.Text.Trim();
string pass = txtPassword.Text.Trim(); if (user == "admin" && pass == "1234") { Form2 menu = new
Form2(); menu.Show(); this.Hide(); } else { lblError.Text = "Invalid username or password."; } }
```

■ Hard-coded credentials are for demonstration only. In production, validate against a database and never store passwords in plain text.

Part 3 — Form2: Main Menu

3.1 Sample Screenshot

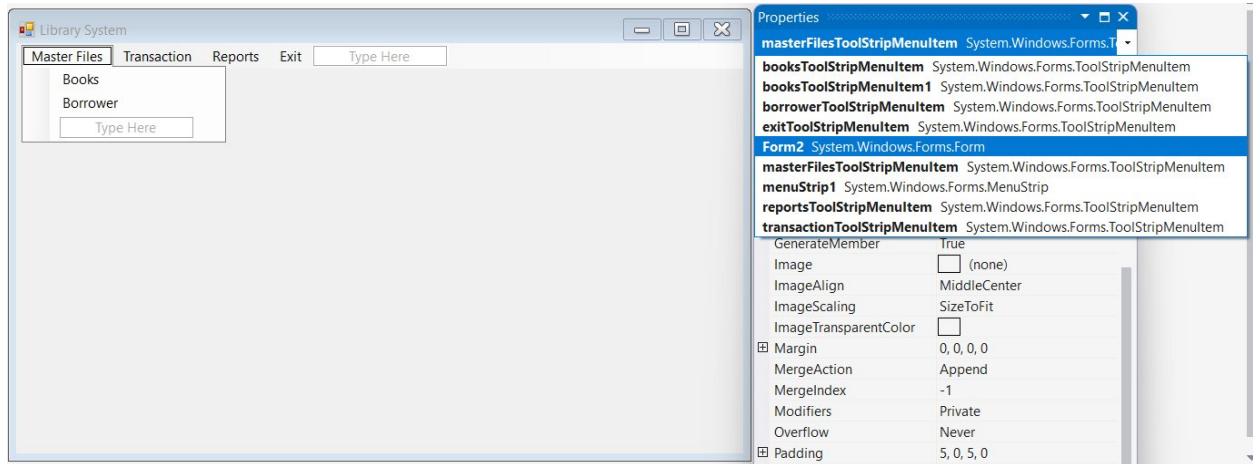


Figure 2 — Form2 with MenuStrip expanded (Master Files > Books, Borrower) and Properties panel

3.2 Add a MenuStrip

Drag a **MenuStrip** from the Toolbox onto Form2. It docks to the top automatically. Build the menu structure below by clicking 'Type Here' in the designer.

Top-Level	Name	Sub-Items (Name → Text)
Master Files	masterFilesToolStripMenuitem	booksToolStripMenuItem → Books borrowerToolStripMenuItem → Borrower
Transaction	transactionToolStripMenuitem	(add sub-items as needed)
Reports	reportsToolStripMenuitem	(add sub-items as needed)
Exit	exitToolStripMenuitem	no sub-items

3.3 Click Event Handlers

```
private void booksToolStripMenuItem_Click(object sender, EventArgs e) { Form3 books = new Form3();  
books.Show(); } private void borrowerToolStripMenuItem_Click(object sender, EventArgs e) { Form4  
borrower = new Form4(); borrower.Show(); } private void exitToolStripMenuItem_Click(object sender,  
EventArgs e) { Application.Exit(); }
```

■ Set **Form2.WindowState** = Maximized for a full-screen main menu.

Part 4 — Form3: Books

4.1 Sample Screenshot

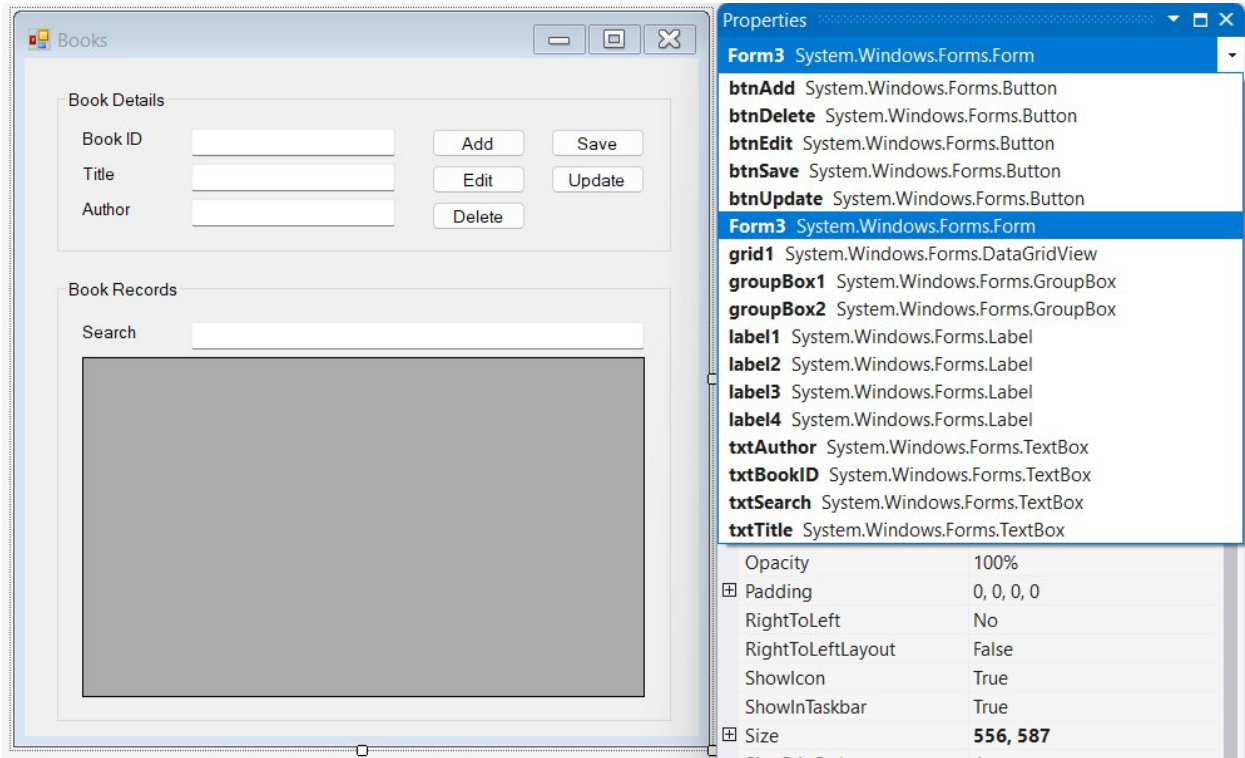


Figure 3 — Form3 Books design view showing groupBox layout, buttons, DataGridView, and Properties panel

4.2 Controls Checklist

Control	Name	Settings
GroupBox	groupBox1	Text = "Book Details"
Label	label1	Text = "Book ID"
Label	label2	Text = "Title"
Label	label3	Text = "Author"
TextBox	txtBookID	Inside groupBox1
TextBox	txtTitle	Inside groupBox1
TextBox	txtAuthor	Inside groupBox1
Button	btnAdd	Text = "Add"
Button	btnEdit	Text = "Edit"
Button	btnSave	Text = "Save"

Button	btnUpdate	Text = "Update"
Button	btnDelete	Text = "Delete"
GroupBox	groupBox2	Text = "Book Records"
Label	label4	Text = "Search"
TextBox	txtSearch	Inside groupBox2
DataGridView	grid1	ReadOnly=True, SelectionMode=FullRowSelect

4.3 CRUD Code

```
private DataTable bookTable; private void Form3_Load(object sender, EventArgs e) { bookTable = new
DataTable(); bookTable.Columns.Add("Book ID"); bookTable.Columns.Add("Title");
bookTable.Columns.Add("Author"); grid1.DataSource = bookTable; SetButtonState("default"); }
private void btnAdd_Click(object sender, EventArgs e) { ClearFields(); SetButtonState("add"); }
private void btnSave_Click(object sender, EventArgs e) { DataRow row = bookTable.NewRow();
row["Book ID"] = txtBookID.Text; row["Title"] = txtTitle.Text; row["Author"] = txtAuthor.Text;
bookTable.Rows.Add(row); ClearFields(); SetButtonState("default"); } private void
btnEdit_Click(object sender, EventArgs e) { if (grid1.SelectedRows.Count == 0) return;
txtBookID.Text = grid1.SelectedRows[0].Cells[0].Value.ToString(); txtTitle.Text =
grid1.SelectedRows[0].Cells[1].Value.ToString(); txtAuthor.Text =
grid1.SelectedRows[0].Cells[2].Value.ToString(); SetButtonState("edit"); } private void
btnUpdate_Click(object sender, EventArgs e) { if (grid1.SelectedRows.Count == 0) return; int idx =
grid1.SelectedRows[0].Index; bookTable.Rows[idx]["Book ID"] = txtBookID.Text;
bookTable.Rows[idx]["Title"] = txtTitle.Text; bookTable.Rows[idx]["Author"] = txtAuthor.Text;
ClearFields(); SetButtonState("default"); } private void btnDelete_Click(object sender, EventArgs
e) { if (grid1.SelectedRows.Count == 0) return;
bookTable.Rows[grid1.SelectedRows[0].Index].Delete(); ClearFields(); } private void
txtSearch_TextChanged(object sender, EventArgs e) { string f = txtSearch.Text.Trim();
bookTable.DefaultView.RowFilter = $"Title LIKE '%{f}%' OR Author LIKE '%{f}%'"; } private void
ClearFields() { txtBookID.Text = txtTitle.Text = txtAuthor.Text = ""; } private void
SetButtonState(string mode) { btnAdd.Enabled = mode == "default"; btnEdit.Enabled = mode ==
"default"; btnDelete.Enabled = mode == "default"; btnSave.Enabled = mode == "add";
btnUpdate.Enabled = mode == "edit"; }
```

Part 5 — Form4: Borrower

Form4 mirrors Form3's layout exactly. Replace Book fields with Borrower fields. The layout, button arrangement, and code pattern are identical — only the field names and DataTable columns change.

5.1 Controls Checklist

Control	Name	Settings
GroupBox	groupBox1	Text = "Borrower Details"
Label	label1	Text = "Borrower ID"
Label	label2	Text = "Name"
Label	label3	Text = "Contact"
TextBox	txtBorrowerID	Inside groupBox1
TextBox	txtName	Inside groupBox1
TextBox	txtContact	Inside groupBox1
Button	btnAdd	Text = "Add"
Button	btnEdit	Text = "Edit"
Button	btnSave	Text = "Save"
Button	btnUpdate	Text = "Update"
Button	btnDelete	Text = "Delete"
GroupBox	groupBox2	Text = "Borrower Records"
Label	label4	Text = "Search"
TextBox	txtSearch	Inside groupBox2
DataGridView	grid2	ReadOnly=True, SelectionMode=FullRowSelect

5.2 Code (adapted from Form3)

```
private DataTable borrowerTable; private void Form4_Load(object sender, EventArgs e) {
    borrowerTable = new DataTable(); borrowerTable.Columns.Add("Borrower ID");
    borrowerTable.Columns.Add("Name"); borrowerTable.Columns.Add("Contact"); grid2.DataSource =
    borrowerTable; SetButtonState("default"); } // All CRUD methods follow the same pattern as Form3.
// Substitutions: // bookTable → borrowerTable // txtBookID → txtBorrowerID // txtTitle →
txtName // txtAuthor → txtContact // grid1 → grid2 private void txtSearch_TextChanged(object
sender, EventArgs e) { string f = txtSearch.Text.Trim(); borrowerTable.DefaultView.RowFilter =
$"Name LIKE '%{f}%' OR Contact LIKE '%{f}%'"; }
```

Part 6 — Navigation Flow

From	Action	Opens	Code
frmLogin	Valid login	Form2	Form2 f=new Form2(); f.Show(); this.Hide();
Form2	Books menu	Form3	Form3 f=new Form3(); f.Show();
Form2	Borrower menu	Form4	Form4 f=new Form4(); f.Show();
Form2	Exit menu	—	Application.Exit();
Form3/4	Close (X) button	Form2	Override FormClosing if re-show needed

■ To re-show Form2 when Form3 or Form4 closes, subscribe to the child's FormClosed event before calling Show():
Form3 books = new Form3(); books.FormClosed += (s, args) => this.Show(); books.Show(); this.Hide();

Part 7 — Build Checklist

Project Setup

- New Windows Forms App (.NET Framework) named LibrarySystem
- Rename Form1.cs → frmLogin.cs (rename all references = Yes)
- Add Form2.cs, Form3.cs, Form4.cs
- Update Program.cs: Application.Run(new frmLogin())

frmLogin

- Controls: label1, label2, lblError, txtUsername, txtPassword, btnLogin
- txtPassword.PasswordChar = '*'
- lblError ForeColor = Red, Text = ""
- btnLogin_Click: validates → shows Form2, hides login

Form2 (Menu)

- MenuStrip: Master Files > Books, Borrower | Transaction | Reports | Exit
- booksToolStripMenuItem_Click → opens Form3
- borrowerToolStripMenuItem_Click → opens Form4
- exitToolStripMenuItem_Click → Application.Exit()

Form3 (Books)

- groupBox1: txtBookID, txtTitle, txtAuthor + btnAdd/Edit/Save/Update/Delete
- groupBox2: txtSearch + DataGridView grid1
- DataTable bookTable bound to grid1 on Form_Load
- CRUD: Add/Save/Edit/Update/Delete all implemented
- Live search via DefaultView.RowFilter in txtSearch_TextChanged

- SetButtonState("default"/"add"/"edit") helper used throughout

Form4 (Borrower)

- groupBox1: txtBorrowerID, txtName, txtContact + 5 buttons
- groupBox2: txtSearch + DataGridView grid2
- Same CRUD + search pattern as Form3, adapted for borrower fields

Common Mistakes to Avoid

■ Not calling **this.Hide()** after showing Form2 from Login — the Login form stays visible behind the menu.

■ Adding rows directly to the DataGridView — always add to the bound DataTable instead; the grid updates automatically.

■ Not converting the angle to radians (applies to Shape project) — and here: forgetting that RowFilter uses SQL LIKE syntax, not C# contains.

■ To connect to a real database (SQL Server / SQLite), replace the in-memory DataTable population with a SqlDataAdapter.Fill(table) call. The grid binding code stays exactly the same.